ADITYA SRIDHAR

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EDUCATION

Vellore Institute of Technology — Vellore, India *2023 – 2027*
B.Tech, Computer Science and Engineering | CGPA 8.21 / 10

PUBLICATIONS & PATENTS

- **Solar-Aware Machine Learning Framework for Short-Term Electricity Demand Forecasting in Delhi Using Weather and Calendar Variables** — Aditya Sridhar (first author). *3rd International Conference on Signal Processing and Computer Vision (SIPCOV-2026), NIT Puducherry, Oct 2026. Presented.*
  5-minute load forecasting on 18 months of Delhi data; solar-availability proxy models the Duck Curve without PV data. Benchmarked 6 models under leakage-free time-aware validation; best reached MAPE 0.77%, R^2 0.9977, tabular learners beat LSTM.
- **Cross-Layer Selective Abstention for Reliable Offline Assistive Perception with Tool-Mediated Voice Interaction for Visually Impaired Users** — Aditya Sridhar (first author). *ICMLDE 2026, conference paper. Under revision.*
- **Indian Patent Application No. 202641110070** — Wearable Navigation Apparatus with Inertially Adaptive Detection and Staged Announcement Scheduling (published, 2026)
- **Indian Patent Application No. 202641115711** — Protected Adaptive Pairwise Sensor Fusion System for Sub-Threshold Compound Hazard Detection and Automated Mitigation (published, 2026)

OPEN-SOURCE CONTRIBUTIONS

- **Kornia** (PyTorch CV library, 11k+ stars) — 4 merged PRs: `focal_loss` NaN gradients for fractional gamma (#5563); `dice_loss` micro mode applied class weights twice (#5564); integer images filtered in their own dtype, so a uint8 Gaussian blur returned zeros (#5669); kernel builders now reject integer dtypes that would truncate or wrap the taps (#5759). Added regression tests.
- **Hugging Face timm** (PyTorch Image Models) — merged fix: non-ASCII class-map labels were silently dropped on Windows; now read as UTF-8 (#2839).
- **deepset Haystack** (LLM / RAG framework) — 2 merged Windows fixes: `EvaluationRunResult` CSV reports crashed on non-ASCII data (#13143); `FileSystemToolResultStore` rewrote line endings, so stored text did not round-trip (#13253).

PROJECTS

BlindAssist — Offline Navigation Aid for Visually Impaired Users | live demo *2026*
Python · PyTorch · YOLOv8 · OpenCV · Flask · Vosk · offline TTS

- Built a fully offline camera-to-speech pipeline (detection → spatial reasoning → decision → speech) announcing obstacle direction and proximity.
- Designed a 3×3 spatial grid with proximity buckets and a path-relevance layer: off-path obstacles stay silent while escalating threats override cooldowns.
- Paired a COCO YOLOv8s detector with a custom YOLOv8n door/dustbin specialist (mAP50 ≈ 0.95), merged per frame by a specialist-wins IoU rule that corrects COCO mislabels.
- 100% direction and proximity accuracy on reviewed announcements at 5.8 FPS on CPU (~0.4 s latency), zero phantom announcements across 7 clips; core logic covered by 61 unit tests.

Offline Voice Assistant — On-Device Speech-to-Speech Desktop Agent | code *2026*
Python · Silero VAD · faster-whisper · Ollama (Llama 3.2 3B) · Piper TTS · tkinter · Gmail & GitHub APIs · OAuth 2.0

- Built a fully local VAD → speech recognition → LLM → speech synthesis assistant with a desktop UI and global hold-to-talk key; ~1.5 s to first spoken word by streaming LLM output into synthesis per sentence.
- Designed a read-only desktop agent: regex fast path plus a constrained-JSON LLM classifier whose action must be backed by the user's words (0 wrong actions on held-out evals); opens apps, explains files, plays YouTube, reads Gmail/GitHub.
- Integrated Gmail (Google OAuth 2.0 + PKCE, gmail.readonly) and GitHub with server-enforced read-only access and encrypted logins; email and web text reach the LLM only as delimited data, never the action classifier; 112 unit tests.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL
- **ML / DL:** PyTorch, transformers, YOLOv8, scikit-learn, LightGBM, self-supervised learning (JEPA, VICReg), LLM agents, forecasting
- **Engineering:** FastAPI, Flask, Streamlit, Docker, REST APIs, SQLite, ONNX / TFLite export, AWS (EC2, S3, IAM), GCP, Git, Linux, pytest

AWARDS & CERTIFICATIONS

- **Amazon ML Challenge 2026** — Top 2% nationally (rank 696 of 35,000+ teams; 89,000+ participants)
- **Code for Life: Hackathon for Everyday Solutions** (VIT, Feb 2026) — Winner, Best Project Award for AI-based anomaly detection in genes
- **Google Cloud Certified — Generative AI Leader** (Jul 2026) | credential
- **Extensive & Reimagined AI (ERA-V3)** (2024–25) — course in deep learning, transformers and LLMs | certificate